

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Emmanuel John
Program/Batch	B083-Networkwalks
Date	20 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability & Responsible Use Disclaimer

All activities documented in this report were performed only against systems, devices, and networks for which I had appropriate authorization or that I personally owned.

The activities were conducted strictly for educational, research, and cybersecurity training purposes. No unauthorized access, exploitation, disruption, or interference was intentionally performed.

The techniques and tools described in this report should only be used within an authorized and controlled environment. Unauthorized access or scanning of computer systems and networks may violate applicable laws and organizational policies.

I accept responsibility for ensuring that any future use of the techniques described in this report is conducted within an appropriate legal and authorized scope.

2. Introduction

This report documents the practical activities completed during **Week 2 of my Cybersecurity & Ethical Hacking Internship with NetworkWalks**.

The week focused on two key areas of ethical hacking and security assessment:

1. **Network Footprinting and Reconnaissance** — gathering publicly available information about an authorized domain using Kali Linux tools.
2. **Network Scanning and Host Discovery** — identifying active devices on my authorized local network using Zenmap.

These activities demonstrate two important stages of a security assessment. Footprinting helps a security professional understand publicly exposed information about a target, while network scanning helps identify active hosts and understand the structure of a network.

For the footprinting exercise, I used **Kali Linux** and several reconnaissance tools. For the network-scanning exercise, I used **Zenmap on Windows** to scan my authorized local network.

Throughout the practical exercises, I documented the commands or configurations used, observations, screenshots, and the potential security relevance of the information identified.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.

curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f** and **DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.1.1** and **WP Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, the connection was refused even after trying several times

I used **Curl** with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint /wp-json/.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified 10 live hosts:

- 192.168.1.91
- 192.168.1.132
- 192.168.1.134
- 192.168.1.166
- 192.168.1.176
- 192.168.1.190
- 192.168.1.191
- 192.168.1.194
- 192.168.1.254
- 192.168.1.185

The example results also included 10 MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

Note: The actual subnet, number of hosts and addresses should be replaced with the results from my own network when submitting the report.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
3	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
4	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
5	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. Review publicly exposed technology information

Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. Keep software updated

CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.

3. Review HTTP headers

HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.

4. Review DNS records regularly

DNS records should be checked periodically to ensure that only required information and services are publicly exposed.

5. Properly configure and monitor the WAF

Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.

6. Perform regular internal network discovery

Organizations should periodically scan their own networks to identify active devices.

7. Investigate unknown devices

Any unexpected device discovered during network scanning should be investigated and verified.

8. Maintain network documentation

Network topology and device information should be documented and updated regularly.

9. Perform security testing with authorization

Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

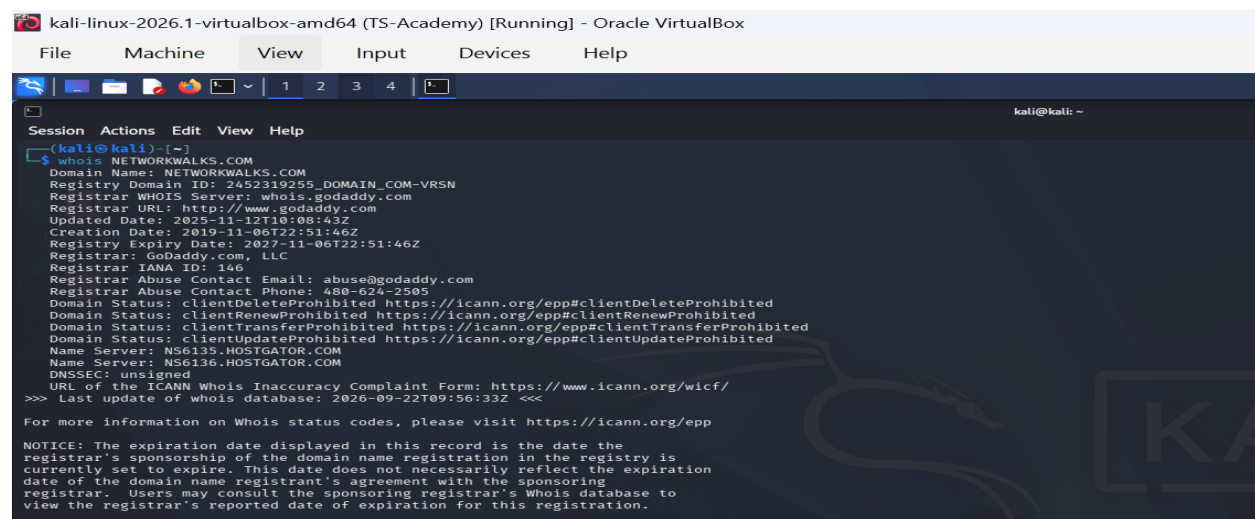
In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected



```
kali-linux-2026.1-virtualbox-amd64 (TS-Academy) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)~$ whois NETWORKWALKS.COM
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-22T09:56:33Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```

```

(kali@kali)-[~]
$ nslookup networkwalks
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
Server:      8.8.8.8
Address:     8.8.8.8#53

** server can't find networkwalks: NXDOMAIN

(kali@kali)-[~]
$ nslookup networkwalks
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
Server:      8.8.8.8
Address:     8.8.8.8#53

** server can't find networkwalks: NXDOMAIN

(kali@kali)-[~]

```

```

(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuer(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/?rel="https://api.w.org/">, <https://networkwalks.com/wp-json/wp/v2/pages/53?rel="alternate">; title="JSON"; type="application/json", <https://networkwalks.com/?rel=shortlink
set-cookie: _wpd_client=49d8b24d031817d8ec456444f11b; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 22 Sep 2026 10:24:25 GMT
server: Apache

```

kali-linux-2026.1-virtualbox-amd64 (TS-Academy) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

kali@kali -

Session Actions Edit View Help

```

(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpd_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[_wpd_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
www.opening: https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.1], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.1], WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidVar2=5051&prVar2=7&pscVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.1]

(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpd_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[_wpd_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.1], Cookies[_wpd_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpd_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.1], WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidVar2=5051&prVar2=7&pscVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.1]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.1], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.1], WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidVar2=5051&prVar2=7&pscVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.1]

```

```

(kali@kali)-[~]
$ wafw00f networkwalks.com

      ?
      ??
      ( " ) ' ; , . ' "
      ( _ ) ; , , ( " ) ; ( , ( ( ; ) " ) "
      / _ _ \ / _ _ \ / _ _ \ / _ _ \ / _ _ \ / _ _ \ / _ _ \ / _ _ \
      \ _ _ / \ _ _ / \ _ _ / \ _ _ / \ _ _ / \ _ _ / \ _ _ / \ _ _ /

~ WAFW00F : v2.3.2 ~
~ Sniffing Web Application Firewalls since 2014 ~

```

```

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

```

```

(kali@kali)-[~]
$

```

Zenmap

Scan
 Tools
 Profile
 Help

Target: 192.168.1.254/24
 Profile: Ping scan

Command: nmap -sn 192.168.1.254/24

Hosts

Services

Nmap Output

Ports / Hosts

Topology

Host Details

Scans

OS

Host

192.168.1.91

192.168.1.132

192.168.1.134

192.168.1.162

192.168.1.166

192.168.1.176

192.168.1.185

192.168.1.190

192.168.1.191

192.168.1.194

dsldevice.lan (192.168.1.254)

nmap -sn 192.168.1.254/24

Starting Nmap 7.991 (<https://nmap.org>) at 2026-09-25 21:19 +0100

Nmap scan report for 192.168.1.91

Host is up (0.33s latency).

MAC Address: 52:BE:E3:90:2A:D3 (Unknown)

Nmap scan report for 192.168.1.132

Host is up (0.54s latency).

MAC Address: 10:68:38:56:AB:13 (AzureWave Technology)

Nmap scan report for 192.168.1.134

Host is up (0.32s latency).

MAC Address: 60:57:18:72:61:EA (Intel Corporate)

Nmap scan report for 192.168.1.166

Host is up (1.1s latency).

MAC Address: FA:99:13:C8:D7:2D (Unknown)

Nmap scan report for 192.168.1.176

Host is up (1.2s latency).

MAC Address: AC:67:5D:27:E9:0D (Intel Corporate)

Nmap scan report for 192.168.1.190

Host is up (1.5s latency).

MAC Address: 16:F7:BF:38:F6:AA (Unknown)

Nmap scan report for 192.168.1.191

Host is up (1.6s latency).

MAC Address: 00:BE:3B:B9:C0:C4 (Huawei Technologies)

Nmap scan report for 192.168.1.194

Host is up (1.4s latency).

MAC Address: 50:78:7D:4D:B7:D4 (Espressif)

Nmap scan report for dsldevice.lan (192.168.1.254)

Host is up (0.0090s latency).

MAC Address: 88:B3:62:43:F2:80 (Nokia Shanghai Bell)

Nmap scan report for 192.168.1.185

Host is up.

Nmap done: 256 IP addresses (10 hosts up) scanned in 25.01 seconds

Viewer

Fisheye

Controls

Legend

Save Graphic

▼ Action

► Interpolation

► Layout

▼ View

☒ address

Navigation

225.0

Zoom

100

Ring gap

30

Lower ring gap

10

e on ring

1.00

with interest factor

3.85

and spread factor

0.57

-End-

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Cybersecurity Intern B083

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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub